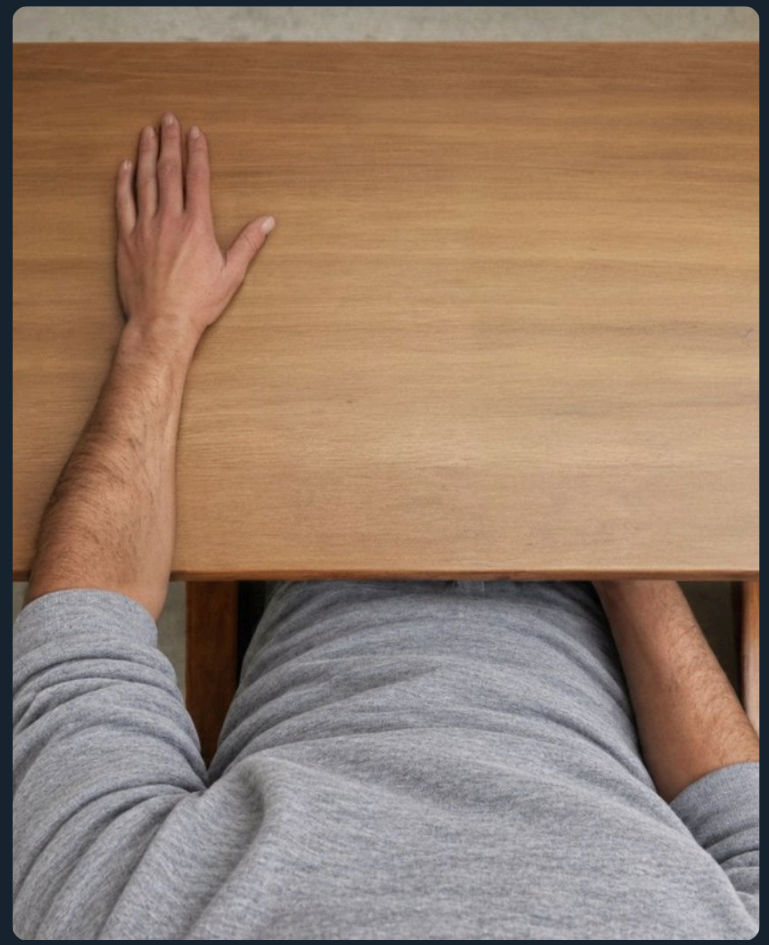
TRDP study · Short version

Can a model that only sees images show the human pattern?

What we did, and what the final test showed

Muhammed Emre Candir

Updated 12 November 2026



The big picture

The study in six steps

1 · Done

Start from a human study

A published experiment gives the pattern to explain.

2 · Done

Make the images

The same hand in several displays, still and moved.

3 · Done

Check the human result

Re-analyse the data and fix one main number.

4 · Done

Test vision models

See if ready-made models already show the pattern.

5 · Done

Build and train our model

3,220 small brain-like networks trained.

6 · Done, 11 November

Final test

Compared the model with humans on unseen images, once.

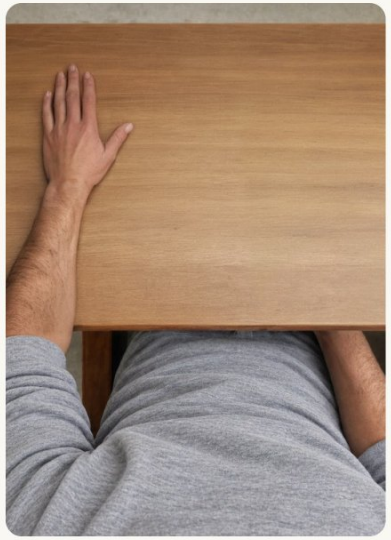
Where we are: the final test was run once. The model passes the rule we fixed in advance, but simpler models pass too (pages 20–21).

Status as recorded on 12 November 2026.

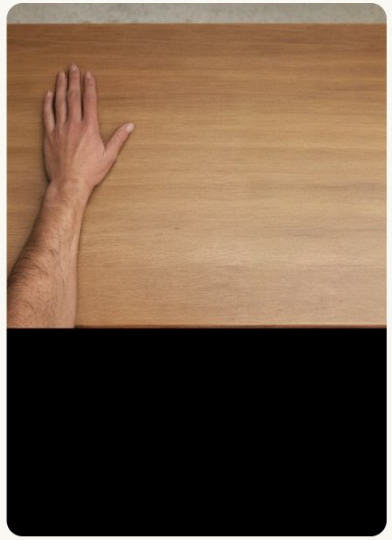
The starting point

The human experiment

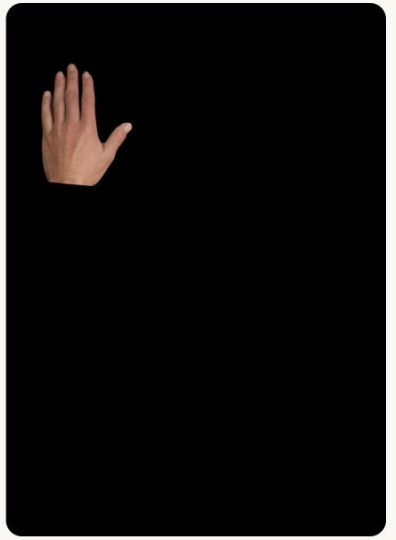
1. 32 people in virtual reality watch a left hand.
2. The index finger either stays still or moves.
3. The hand is shown in three displays.
4. A magnetic pulse (TMS) is given to the brain's movement area.
5. The size of the muscle's response (MEP) is measured.



Full body



Upper limb

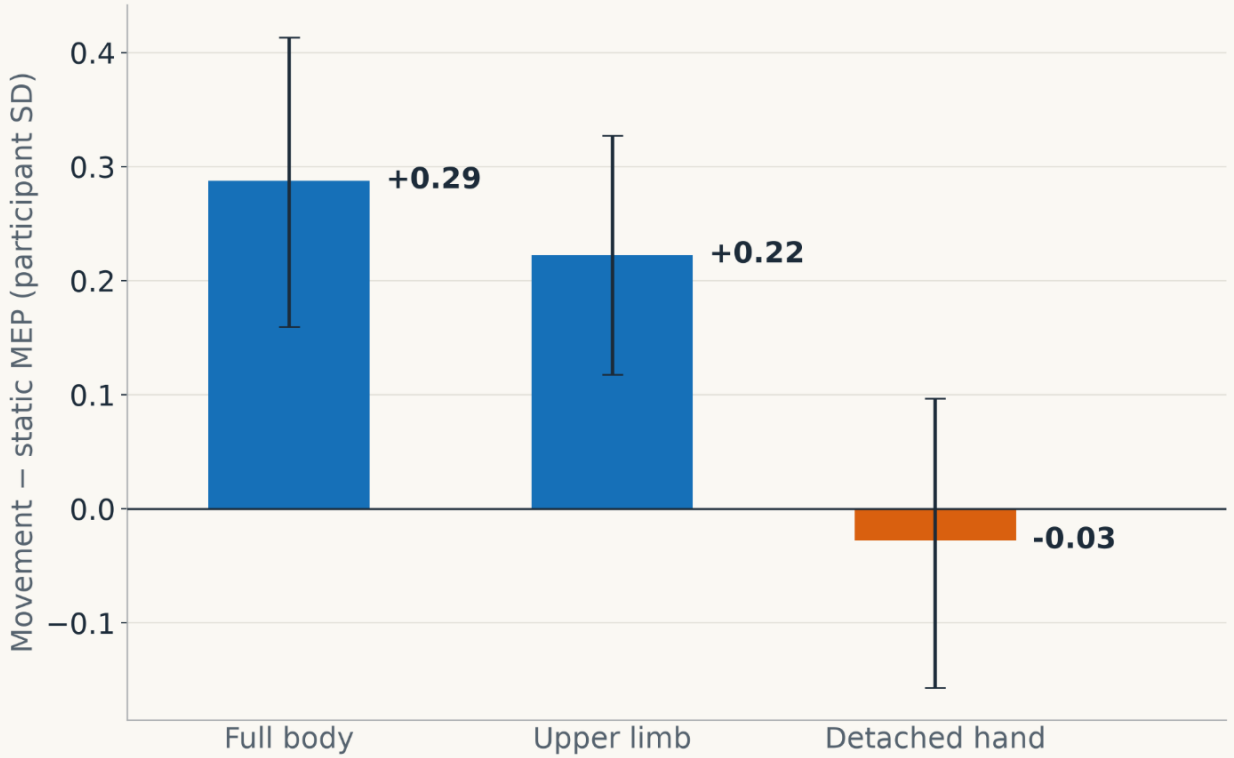


Detached hand

Terms: TMS = transcranial magnetic stimulation, a magnetic pulse to the brain. **MEP** = motor evoked potential, the muscle's response to it.
Images are our own generated versions. Study: Matamala-Gomez, Frisco et al. 2025, Scientific Reports.

The starting point

What humans show



Hand connected to a body: seeing the finger move makes the muscle response larger.

Detached hand: no clear change.

Terms: FDI = first dorsal interosseous, the index-finger muscle. **SD** = standard deviation, the usual spread between people. **CI** = confidence interval, the likely range.

Source: our re-analysis of the study's public data.

The starting point

Our question, and its limits

Can a model that only sees images reproduce this pattern, and which parts does it need to do so?

What we can say

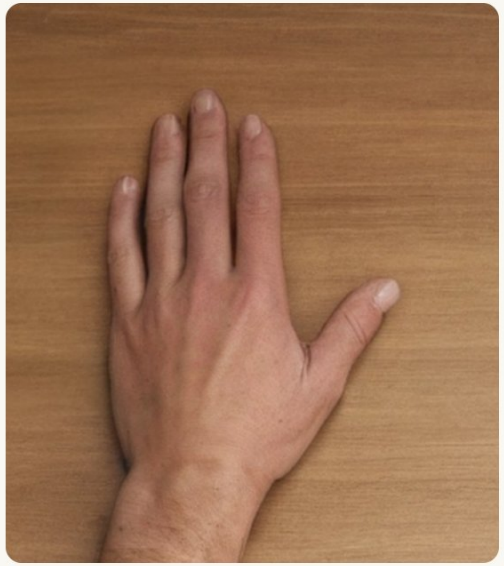
- Whether this model shows the pattern on images it has not seen
- Which parts it needs

What we cannot say

- That we repeated the human experiment
- Anything about feeling ownership or control
- That the model’s signal is a real muscle response
- Anything beyond our 20 hands

The images, in brief

Same hand; the finger movement is the only change



Still



Moved

- **20 AI-made hands**, so only 20 real movements.
- The movement is **calculated**, not drawn by AI.
- **2,880 images** in different looks, plus 960 detached-hand images.
- Extra **control displays** help separate “connected” from “how much is visible”.

All images were accepted by one reviewer who knew the study’s aims. No human data exists for the control displays.

Checking the human result

We re-analysed the data and chose one main number

18 / 18

readable published values reproduced, from 11,280 muscle responses.

0.283

Main number: connected displays minus detached hand. Likely range 0.147 to 0.419.

24 / 32

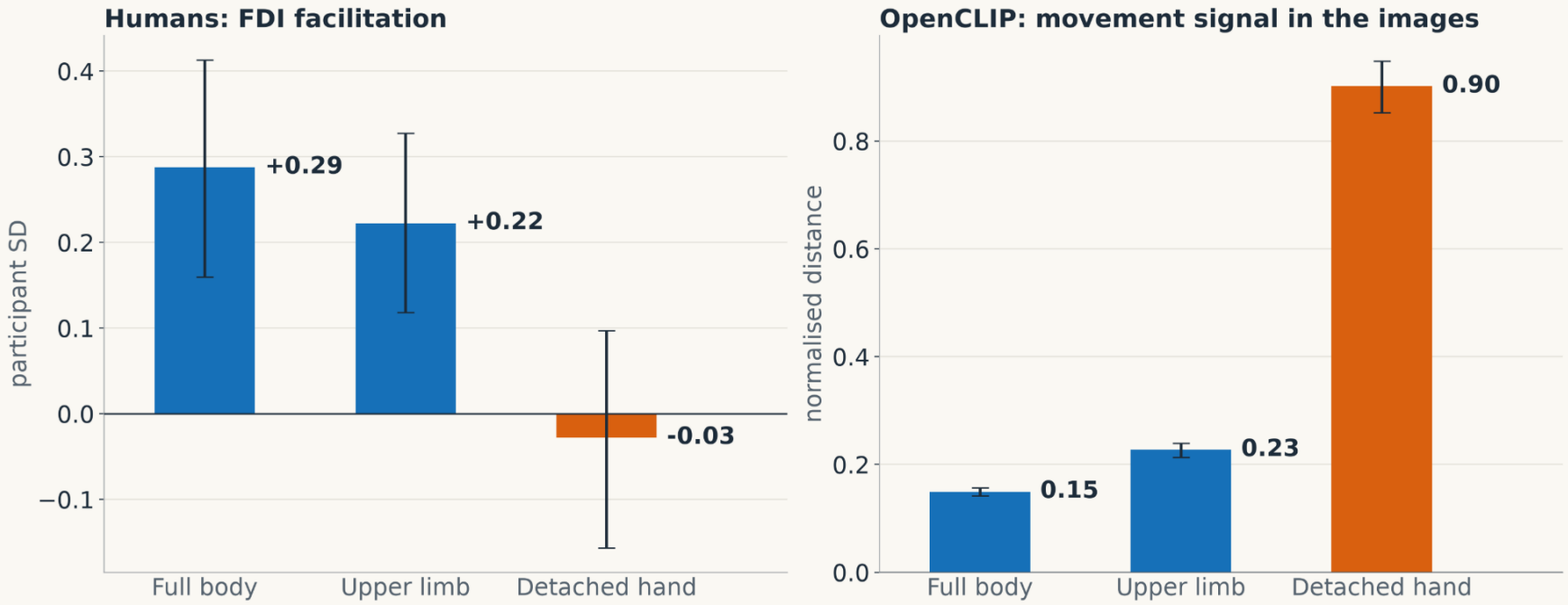
people show this difference in the expected direction.

- We count **each person once**. Some published results are weaker this way.
- The human data cannot tell if **connection** or **the amount of visible body** causes the effect.
- The target and its rules were **fixed on 22 October**.

Units: standard deviations (the usual spread). The number is written C-FDI in the long deck. Source: analysis spec; public data on OSF (Open Science Framework).

First test: ready-made vision models (Gate 1)

Vision models show the opposite of humans



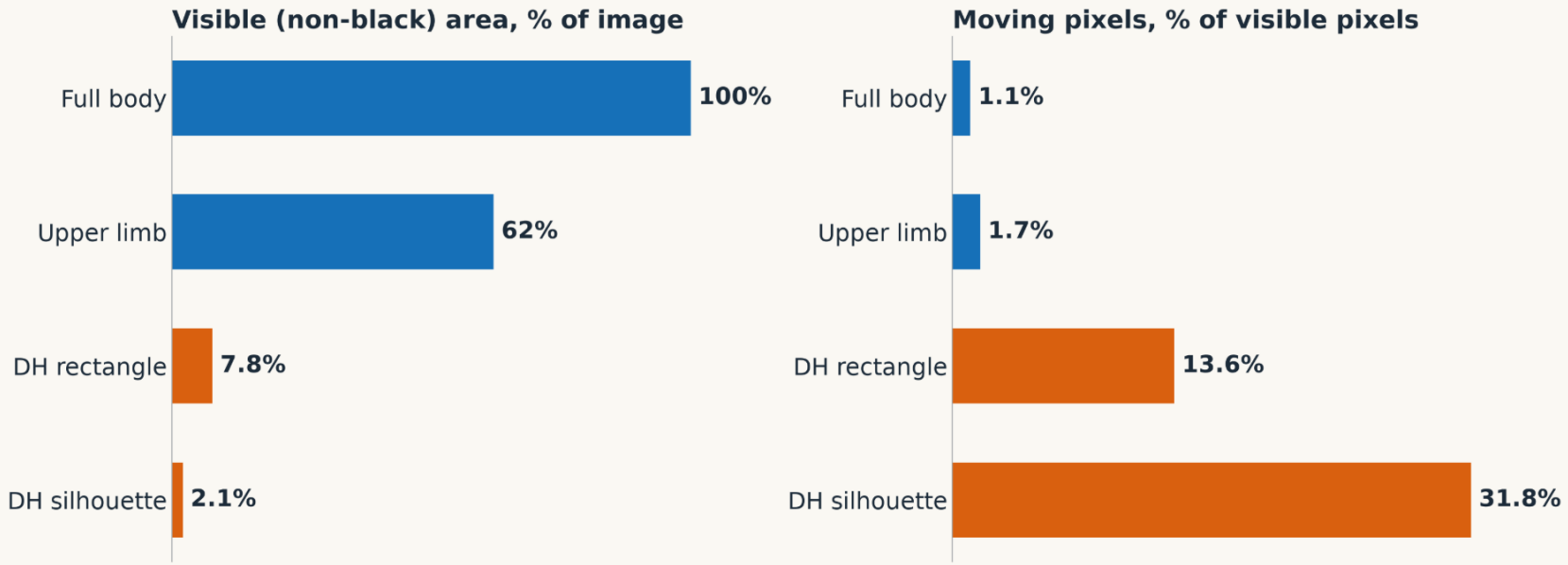
0 of 20 hands follow the human direction. Across 76 other ways of analysing, **none** gives the human direction.

Terms: Gate 1 = our first checkpoint. **OpenCLIP**, **DINOv2** = ready-made vision models. **Movement signal** = how much the model's numbers change when the finger moves.

The panels use different units: compare the shape, not the height. No training at this stage. 10,560 images.

First test: ready-made vision models (Gate 1)

Why: with less to see, the finger is a bigger part of the picture



- Against black, the moving finger changes the image **about 3 times more**.
- Zooming in makes the reversal smaller but does not remove it.

Terms: Rectangle = the earlier detached hand with a piece of desk. **Silhouette** = only the hand on black, now our main detached hand.
Measured on 80 image families. Source: baselines document.

First test: ready-made vision models (Gate 1)

What this first test tells us

The test means something

The pattern is not already in the image signal. A model has to add something to get it.

“Attached” is not given

Ready-made OpenCLIP numbers do not carry over “connected to an arm” to new displays.

The limits

20 generated hands, two vision models, and a visual signal is not a muscle response.

Gate 1 was passed on 23 October, so we could move on to the model.

The model we built

From an image to a simulated muscle response

1

Image

The hand in one display.

2

Vision model

OpenCLIP, not changed by training, turns it into numbers.

3

Spiking layer

256 neurons that send spikes, like brain cells.

4

Gate

Controls how much signal passes on.

5

Motor neurons

64 for the index muscle, 64 for the little-finger muscle.

The virtual pulse: we hit the motor neurons with a simulated pulse, as in humans, and count the extra spikes. That count is the model's "muscle response".

OpenCLIP is the main vision model because it is the least favourable one in the first test. The model is never told which display it sees.

The model we built

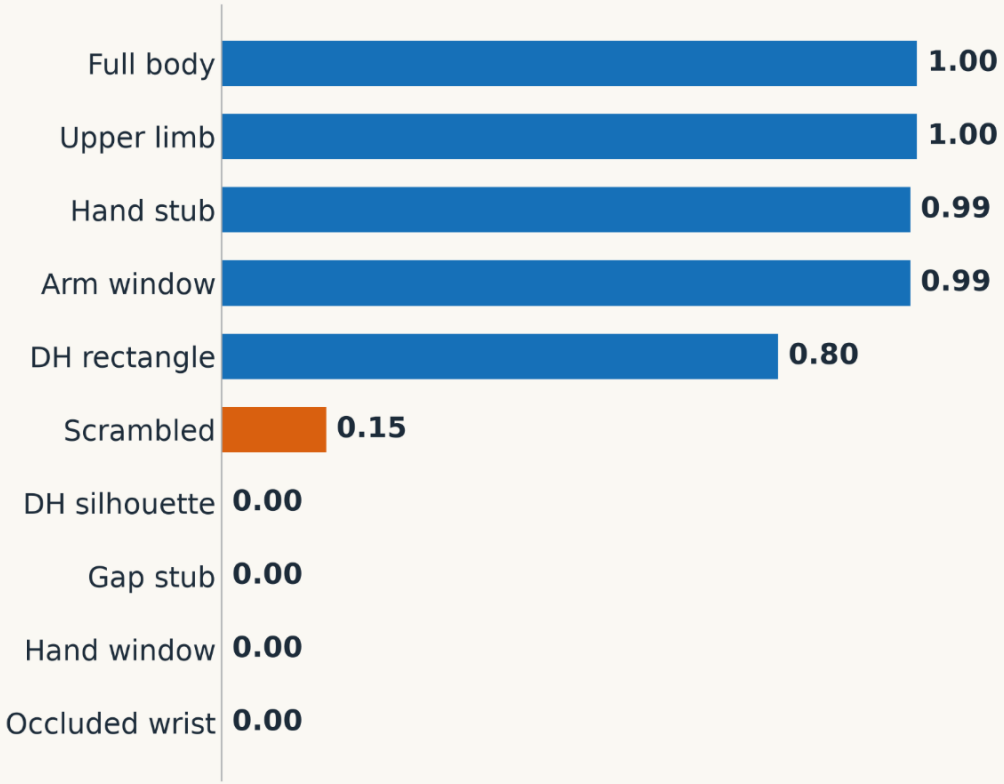
Three ideas to compare

H0, hand only: only the hand counts.

H1, image change: more change in the image, more drive. Nothing is fitted.

H2, continuity gate: the gate opens when the forearm continues at the wrist.

H2 reads continuity from the image. One number is then fitted to the human result, on training hands only.



Terms: H = hypothesis. **Continuity** = how well the forearm carries on at the wrist, from 0 to 1. **Stub, gap, window** = control displays.
We also compare a model with no gate, learned gates and a non-spiking copy. Source: analysis spec; model-method document.

The model we built

How the model is trained

What it learns

- One task: detect that the index finger moved
- From full-body images only
- Motor neurons stay quiet, near rest, because it only watches

What it never gets

- Muscle-response data
- Labels saying which display it sees
- The kept-aside hands

So any display pattern must come from the model’s design, not from being trained on it.

Assumption: the little-finger response is held about equal for still and moved images. Near rest = about 1 spike per second, at most 3. Source: model-method document.

The model we built

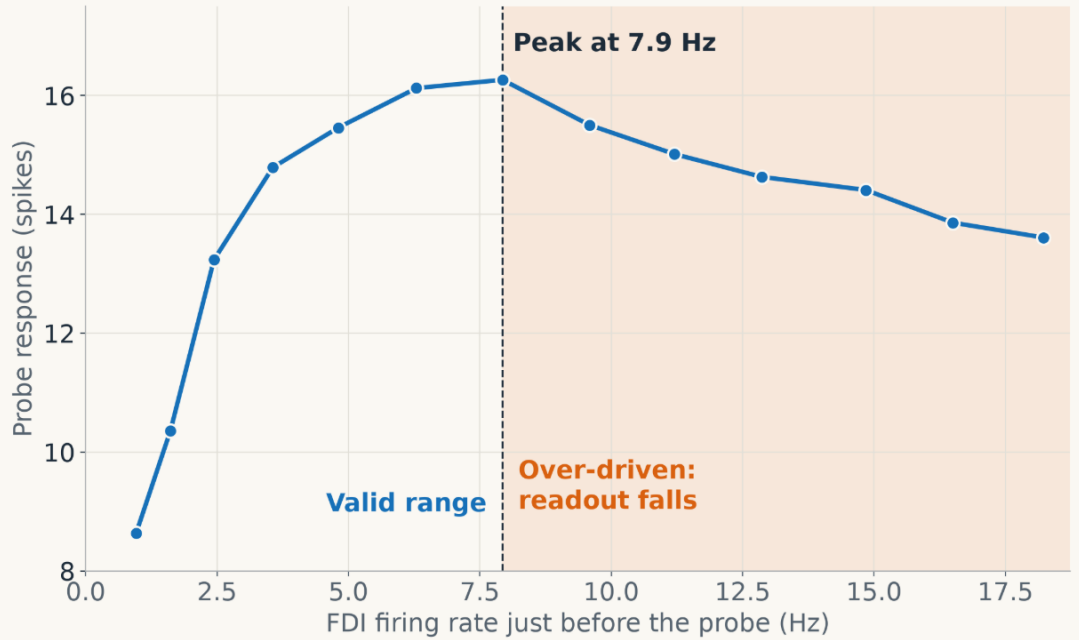
The first version failed; we fixed it on training data

Before: no larger response to movement (−0.47 spikes). Neurons fired too much.

Two repairs: keep motor neurons near rest, and make all neurons in a group equal.

After: a clearly larger response (+2.40 to +6.57 spikes).

Remaining limit: the readout falls when neurons fire too much. Real muscles do not do this.



Terms: Readout = the extra spikes after the pulse. **Hz** = spikes per second. **MEP** = the muscle's response.
No fix used results from other displays or the kept-aside hands. Source: model-method document.

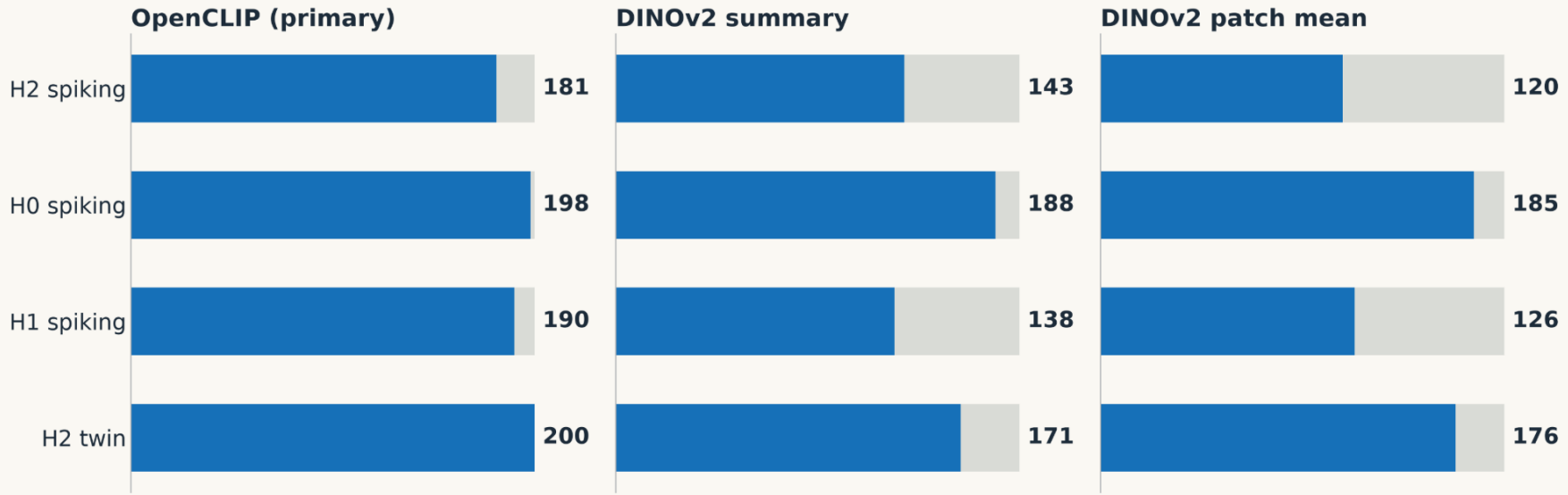
Training the networks

We trained 2,400 networks; 2,016 pass all health checks

12 model types × 20 data splits × 10 random starts

Each split: 18 hands to train, 1 to check, 1 kept aside and never used

Six lab GPUs at CREMI, from 25 to 26 October



Terms: Health check = does the network behave sensibly? It does not compare with humans. **GPU** = graphics processor. **CREMI** = University of Bordeaux lab computers.

Each model type has 200 networks. Training-side data only. Source: validation; model-method document.

Training the networks

384 networks failed; we kept them and set a rule

214
fire too much before the pulse (“over-driven”)

145
learn the task, but it does not carry over to the check hand

25
pass the rest but show no clear larger response to movement

The rule, set before any evaluation

- No network is thrown away for these failures.
- Over-driven results are analysed **two ways**. A test counts as passed only if both pass.
- An unclear result never counts in favour of an idea.

No training setting was changed after these failures. Source: model-method document; validation.

After training

More checks, then everything was frozen

Met

Gate 2

The model is stable, its response to the pulse is smooth, and the three ideas predict differently.

8–10 November

820 more networks

Loops in the visual layer, nine other settings, training on all displays, a finer time step.

10 November

Independent review

18 points found and the claim narrowed. No setting was changed.

11 November

Freeze

Code and inputs locked by digital fingerprints before the final test.

- **One surprise in Gate 2:** the gate's fitted number is 0 in almost every network. The human detached-hand effect is slightly below zero, and a gate can at most close fully.
- **Rehearsal:** all 3,220 networks went through the test code on training hands first; one fault was found and fixed.

Terms: Gate 2 = our second checkpoint. **Freeze** = nothing may change after this point. **Loops** = connections inside the visual layer ("recurrent").
Training hands only. Source: model-method document; independent review.

After training

Even without a gate, the trained model weakens the detached hand

Trained on full-body images only

0.49

detached-hand response, as a share of the full-body response

Trained on all three displays

1.14

the detached hand is no longer weaker

What this means: the human-like order partly comes from what the model was trained on. With experience of every display, the gate is needed to get the human pattern.

Terms: Without a gate = the same trained network with the gate always open. Humans: about −0.10.
Found by the independent review; training hands only, before the final test. Source: model-method document.

The final test

What counted as success for the gate model (H2)

Built in, or seen before the test

- Main number above zero
- Larger response in full body
- Detached hand weak, as in humans
- Beats the hand-only model
- Beats a “visible area” model on the controls

The real tests

- Larger response in the upper limb too
- Upper limb and full body in the human ratio
- The same with other vision models and settings

Even a full pass would not show how the brain gets such a signal, or that humans use continuity and not visible context.

Tests fixed in advance; which ones were really open was recorded on 10 November, before the test. Source: analysis spec; criteria status.

The final test

The model passes the rule we set in advance

0.96

Upper-limb response as a share of full body. Humans: 0.77 (likely range 0.34 to 1.78).

3,220

networks tested once on hands they had never seen; 0 failures.

13 / 13

other model versions agree; one reading of one vision model is unclear.

- The **upper limb** was the only open test for H2, and it passed.
- What the model learned from full bodies carries over to the arm alone.
- Run once on 11 November, on six lab computers, and scored once.

Model values are in the model's own units: compare ratios, not sizes.
Source: model-method document, held-out result; docs/evaluation.

The final test

What the result shows, and what it does not

What it shows

- The model gives the human pattern on unseen hands
- The upper limb carries over at a human-like ratio
- This holds across vision models and settings

What it does not show

- That the gate does it: the model without a gate carries over the same way
- That attachment is the reason: a “visible area” gate and simple image change also pass
- Anything about the brain or the feeling of ownership

Only the control displays tell the ideas apart: there, only the gate model predicts a difference on both matched pairs. No human data exist for them, so a human experiment on these images is the clear next step.

The control comparison across all models was looked at after the test (exploratory). Source: model-method document.

Where we stand

Everything up to the final test is done

Done

- Images and control displays
- Human target
- Gate 1
- 3,220 networks trained
- Gate 2 and the freeze
- The final test

Found along the way

- The gate closes fully (c = 0)
- No gate is needed for the main order
- The paper’s displays cannot separate the ideas

Still to do

- Project report
- Paper-style report
- Hand-in (assumed 29 January 2027)

Terms: c = the gate’s one fitted number; 0 means fully closed for a detached hand.
As recorded on 12 November 2026. Source: validation; final-phase plan.

Where we stand

The weak points we know about

- **Only 20 generated hands**, from one image generator.
- **One reviewer**, who knew the aims, accepted all images; two other people later agreed, informally.
- **Most tests for H2 were built in**; only two were really open.
- **The readout falls** when neurons fire too much; real muscles do not.
- **The paper’s three displays** cannot separate attachment from visible area.
- **No working spinal stage** in the model.

49
major decisions reviewed

41
references checked

Spinal stage = a spinal-cord step between brain and muscle. Source: decision audit; independent review.

What comes next

From here to the report

December

Write the project report from the frozen results.

December–January

A paper-style article on the same results.

29 January 2027

Hand-in (assumed date).

Open question

A human experiment on the control displays would settle it.

In short: humans show a pattern, ready-made vision models show the opposite, and our trained model reproduces it on unseen hands. On the paper’s displays, simpler ideas reproduce it too.